

Inadequacy of classical physics

Classical mechanics is a branch of physics that deals with the motion of macroscopic bodies or objects (i.e. **the size range greater than 10^{-8} m**) under the influence of forces. While it was groundbreaking when first developed by Sir Isaac Newton in the 17th century, it has certain limitations that were discovered over time. In this answer, we will discuss these limitations in more detail:

1. Classical Mechanics is not applicable to extremely small objects: Classical mechanics assumes that particles have a definite position and momentum, which is not true in the quantum world. This limitation became apparent in the early 20th century with the discovery of quantum mechanics. Quantum mechanics is a branch of physics that deals with the behavior or motion of particles on an atomic and subatomic level (i.e. the size range is in between 10^{-8} m to 10^{-15} m) i.e. microscopic particles. It has been successful in explaining phenomena such as the photoelectric effect, blackbody radiation, and the behavior of electrons in atoms, which cannot be explained by classical mechanics.

2. Classical Mechanics is not applicable to objects moving at very high speeds: Classical mechanics assumes that the speed of an object can be infinite, it is not true in the relativistic world. The theory of relativity which was developed by Albert Einstein in the early 20th century, explains the behavior of objects moving at high speeds (i.e. equal to the speed of light). The theory of relativity has been successful in predicting phenomena such as time dilation, length contraction, and the equivalence of mass and energy.

3. Classical Mechanics can not well explain the behavior of systems with many particles: Classical mechanics is not well suited for dealing with systems that have many particles. This is because it is difficult to solve the equations of motion for systems with many particles, and the behavior of the system can become chaotic. The theory of statistical mechanics, developed in the late 19th century, addresses this limitation by using probability distributions to describe the behavior of large systems.

4. Classical Mechanics can not explain the behavior of objects that are very far apart or have very high masses: Newton's law of gravity works well for objects that are close together, but it fails to explain the behavior of objects that are extremely far apart or have very high masses, such as black holes and galaxies. The theory of general relativity, developed by Einstein in the early 20th century, provides a better explanation of the behavior of objects with very high masses and gravitational fields.

In summary, while classical mechanics is still useful for understanding the behavior of macroscopic objects, its limitations have led to the development of new theories, such as quantum mechanics, relativity, statistical mechanics, and chaos theory, which can explain the behavior of the universe at different scales and levels of complexity.



Classical mechanics is the study of the motion of everyday objects in accordance with the general principles first developed by Newton with later modifications by Einstein.



Quantum mechanics is a set of mathematical principles that attempts to explain the behavior of atoms and sub-atomic particles.

Need for quantum physics

- ✓ Quantum theory is needed because many phenomena at the microscopic level can not be explained using classical theory, eg: Photoelectric effect, interaction among elementary particles
- ✓ Quantum mechanics, science dealing with the behaviour of matter and light on the atomic and subatomic scale. It attempts to describe and account for the properties of molecules and atoms and their constituents-electrons, protons, neutrons, and other more esoteric particles such as quarks and gluons.
- ✓ Quantum mechanics led to the development of things like lasers, light-emitting diodes, transistors, medical imaging, electron microscopes, and a host of other modern devices. Your cell phone would not exist without the science of quantum mechanics
- ✓ Five practical uses of quantum mechanics?
 - MRI SCANNERS FOR MEDICAL IMAGING. Magnetic resonance imaging (MRI) has revolutionized how we diagnose diseases, providing medical professionals with a non-invasive method to generate images inside human bodies. ...
 - LASERS. ...
 - SOLAR CELLS. ...
 - ELECTRON MICROSCOPES. ...
 - Global Positioning System (GPS)

Need for quantum physics

What is the everyday use of quantum physics?

1: Toaster: Toasters are generally referred to as the reason why Quantum Physics came into existence. The rod in the toaster gets hot, which, in turn, is responsible for toasting the bread.

2: Fluorescent light : The light which you are getting from the tubes or those curly bulbs is a result of a quantum phenomenon only. In fluorescent lighting, a small amount of mercury vapour is excited into the plasma. Mercury has the ability to emit light in the visible range. So, the next time you switch on the lights of your room at night, make sure you thank Quantum Physics.

3. Computer & Mobile Phone; The whole computer world is based on the principle of Quantum Physics. Quantum Physics talks about the wave nature of electrons, and, so, this forms the basis of the band structure of solid objects on which semiconductor-based electronics are built. Not to forget that we are able to manipulate the electrical properties of silicon only because we can study the wave nature of electrons. Once the band structure is changed, the conductivity alters as well. How can the band structure be changed? Of course, Quantum Physics knows the answer!

Need for quantum physics

4. Laser: The principle on which laser works is based on Quantum Physics. The working of lasers involves spontaneous emission, thermal emission, and fluorescence. An electron, when excited, will jump to a high-energy level. However, it will not stay in the high-energy level for a long time, and jump back to the lower energy state which is more stable; and, thereby, emit light. The quantum mechanical state of the atom is also affected by external photons which are at a frequency associated with the atomic transition.

Historical overview of Quantum Mechanics

1. Founder of Quantum Mechanics

Quantum mechanics began in 1900 when German scientist Max Planck published a paper on the effect of radiation on a black substance. Planck assumed that, like matter, energy is made up of separate units rather than a continuous electromagnetic wave during his investigations. Quanta were the name given to these tiny particle units. This hypothesis could explain natural phenomena such as how heat behaves on solids and how light is absorbed at the atomic level.

2. Quantum Mechanics Theory

Quantum theory is the foundation of modern physics, explaining the nature and behaviour of matter and energy at the atomic and subatomic levels. Quantum mechanics and quantum physics are terminologies that describe the nature and behaviour of matter and energy at that level. Quantum computing, which employs quantum theory to dramatically boost computing capabilities beyond what is conceivable with today's classical computers, has attracted major funding from a number of countries.

3. Quantum Theory Development

1. Planck assumed that energy was made up of discrete units, or quanta, in 1900.
2. In 1905, Albert Einstein proposed that radiation, as well as energy, be quantized in the same way.
3. Louis de Broglie proposed in 1924 that there is no fundamental difference in the makeup and behaviour of energy and matter; both can act as if they are formed of particles or waves at the atomic and subatomic levels. This theory became known as the principle of wave-particle duality, which states that elementary particles of both energy and matter act as either particles or waves depending on the circumstances.
4. Werner Heisenberg claimed in 1927 that precise, simultaneous measurement of two complementary quantities – such as a subatomic particle's position and momentum – is impossible. Their simultaneous measurement, contrary to classical physics principles, is inherently erroneous; the more precisely one value is measured, the more flawed the measurement of the other value will be. The uncertainty principle was born out of this idea, prompting Albert Einstein's famous remark, "God does not play dice."

4. What are the main features of quantum mechanics?

Its two most important features have been preserved in modern quantum mechanics. They are:

- (1) The existence of stationary, non-radiating states.
- (2) The link between the energy difference between the initial and final states in a transition and the frequency of radiation

5. Significance of Quantum Mechanics

Quantum mechanics enables the calculation of physical system attributes and behaviour. Molecules, atoms, and subatomic particles are examples of microscopic systems that are usually described with this term.

Atomic clocks, the world's most exact clocks, can measure time using quantum theory principles. They keep track of the precise radiation frequency required to cause electrons to hop between energy levels.